

FoodHelper Service

CHG-MACRO-001 — v2

Macronutrient & Fiber Range Changes — Developer Change Document

Change ID	CHG-MACRO-001 v2
Supersedes	CHG-MACRO-001 v1 — substantial corrections applied
Date	May 2026
Client doc ref	Macronutrient_Range_Review_v230526f-update24May.docx
Stage	Stage 1 only — keto (Stage 2) excluded
Backend changes	SetDefaultsOfUserGuidance (fiber only), CaloryBrakets (fats), both algorithm SPs (fat deviation penalty)
Frontend changes	Dietitian macro panel — protein dropdown ceiling, fats dropdown extension + notification, carbs vegan exception, fiber cumulative dropdown
Algorithm SPs	Fat deviation penalty ONLY — no other SP changes

THIS IS A CORRECTED DOCUMENT — READ BEFORE PROCEEDING

v1 contained three fundamental errors confirmed by client review:

1. PROTEIN: direction reversed. Values ABOVE ceiling are greyed out, not below. Protein change is FRONTEND ONLY.
2. CARBS: field is NOT read-only. Dropdown unchanged except vegan 2400-2800 kcal (30% greyed out only).
3. FIBER: cumulative availability logic, not a simple floor grey-out. Full dropdown is 15-18-21-23-25-30-33-35-38-40g.

Discard v1 entirely. Use this document only.

1. Architecture Overview

The macro values in the dietitian panel are driven by UserNutritionsByDietitian, which SetDefaultsOfUserGuidance writes at profile setup. The algorithm reads grams from that table. The dietitian panel displays percentages converted from those grams and writes updated grams back when the dietitian makes a change.

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KEY SCOPE POINT
Protein ceiling: FRONTEND ONLY. The 1.2g/kg default is safe within all ceiling ranges even at 200kg.
No backend SP change is required for protein. The SP does not need to enforce the protein ceiling.
Carbs: FRONTEND ONLY (one exception only — vegan 2400-2800 kcal).
Fiber floor: BACKEND REQUIRED. Dynamic floor must be written to DB so frontend knows what to enable.
Fats: BOTH backend (CaloryBrackets) and frontend (dropdown + notification) and algorithm SP (deviation penalty).

2. Change 1 — Protein Ceiling (Frontend Only)

⚠

CRITICAL CORRECTION FROM v1
The protein matrix defines an UPPER LIMIT (ceiling), not a lower floor.
Values ABOVE the ceiling are greyed out. Values AT OR BELOW the ceiling are selectable.
The dietitian selects DOWNWARD from the ceiling, not upward from a floor.
This is a FRONTEND-ONLY change. No SP change required.

2.1 Diet type detection (frontend)

The diet flags are already available in the patient API response. Use the same two-flag logic the backend already uses:

- IsVegan = true → apply vegan ceiling column
- IsVegetarian = true (and IsVegan = false) → apply vegetarian ceiling column
- Neither flag → apply omnivore ceiling column (default)

No new API call or new flag needed.

2.2 Protein ceiling matrix

The ceiling is the MAXIMUM value the dietitian can select. All values above the ceiling are greyed out (visible, not clickable, cursor: not-allowed).

Kcal tier	Omnivore ceiling	Vegetarian ceiling	Vegan ceiling
1,200 – 2,000 kcal	35% (no change)	30%	25%
2,100 – 2,300 kcal	32.5%	30%	25%
2,400 – 2,800 kcal	30%	27.5%	22.5%

2.3 Exact greyed-out values per profile

Current full dropdown: 10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30, 32.5, 35. No new values added.

Profile	Selectable (active)	Greyed out
Omni 1,200–2,000	10 → 35 (all — no change)	—
Veg 1,200–2,300	10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30	32.5, 35
Vegan 1,200–2,300	10, 12.5, 15, 17.5, 20, 22.5, 25	27.5, 30, 32.5, 35
Omni 2,100–2,300	10 → 32.5	35
Omni 2,400–2,800	10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30	32.5, 35
Veg 2,400–2,800	10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5	30, 32.5, 35
Vegan 2,400–2,800	10, 12.5, 15, 17.5, 20, 22.5	25, 27.5, 30, 32.5, 35

2.4 Frontend implementation spec

Item	Detail
Data source	IsVegan, IsVegetarian flags + EnergyTarget (daily kcal) — both already in patient API response
Ceiling lookup	Hardcode the 3×3 matrix above in frontend logic. No new API call.
Greyed-out style	Visible, not clickable, cursor: not-allowed, colour: #AAAAAA or equivalent disabled state
Tooltip on hover	'Not available for this patient profile'
On profile load	Apply ceiling immediately based on patient profile flags and kcal. Do not wait for interaction.
Backend SP change	NONE REQUIRED for protein.

3. Change 2 — Fats (Backend + Frontend + Algorithm SP)

3.1 Fat ceiling: 35% → 45% — Backend

Update CaloryBrackets.FatsMax to 45% equivalent grams across all rows. SetDefaultsOfUserGuidance reads FatsMax from CaloryBrackets and writes it to UserNutritionsByDietitian. The algorithm then reads

@FatMax from UserNutritionsByDietitian. No SP logic change needed — just the data values in CaloryBrakets.

Kcal tier	Old FatsMax (35%)	New FatsMax (45%)	Change in grams
1,200 kcal	46.7g	60.0g	+13.3g
1,500 kcal	58.3g	75.0g	+16.7g
2,000 kcal	77.8g	100.0g	+22.2g
2,600 kcal	101.1g	130.0g	+28.9g
2,800 kcal	108.9g	140.0g	+31.1g

3.2 5% deviation penalty — Both algorithm SPs

The client document (section 6.3) requires fats not to deviate more than 5 percentage points from the profile set value. The existing scoring penalty handles over-target fat but not deviation from the set value. Add the following penalty term to the ORDER BY scoring in all three meal sections (breakfast, lunch, dinner) in BOTH algorithm SPs. The scoring weights (1.5 / 15 / 12 / 3) do not change.

⚠ BOTH SPs REQUIRE THIS ADDITION

Apply identically to AlgorithmCalculationsV4 AND LIVE_version_optimized.

Add to breakfast, lunch, AND dinner scoring sections in each SP.

Scoring weights (1.5 / 15 / 12 / 3) are unchanged.

```
-- Add at top of SP with other macro variable declarations:
DECLARE @ProfileFatPct FLOAT;
SET @ProfileFatPct = ((@FatMin + @FatMax) / 2.0) * 9.0 / @DailyCalories;
-- FatMin/FatMax in grams; multiply by 9 kcal/g; divide by daily kcal = fat % as decimal

-- Add to ORDER BY for each meal (breakfast/lunch/dinner):
-- Keep existing over-target penalty:
+ CASE WHEN r.NfTotalFat > @[Meal]TargetFat
      THEN (r.NfTotalFat - @[Meal]TargetFat) * 10 ELSE 0 END

-- ADD deviation penalty (new):
+ CASE WHEN r.NfCalories > 0
      AND ABS((r.NfTotalFat * 9.0 / r.NfCalories) - @ProfileFatPct) > 0.05
      THEN ABS((r.NfTotalFat * 9.0 / r.NfCalories) - @ProfileFatPct) * 200
      ELSE 0 END
```

3.3 Fat dropdown extended + safety notification — Frontend

Add four new values to the fat dropdown: 37.5%, 40%, 42.5%, 45%.

Current fat dropdown	New fat dropdown
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20, 22.5, 25, 27.5, 30, 32.5, 35

20, 22.5, 25, 27.5, 30, 32.5, 35, 37.5, 40, 42.5, 45

Safety notification — display when fat % is 42.5% or above:

Trigger	Detail
On selection	Appears immediately when dietitian selects 42.5% or above in dropdown
On page load	Also visible immediately if profile is already set at 42.5% or above when dietitian opens the panel
Dismissal	Disappears if dietitian selects below 42.5%. Does NOT block saving.
Position	Red banner at top of the macro setup panel

⚠ English notification text (confirmed by client)

Safety check: review for contraindicating comorbidities before activating — this plan can exceed 45% fat.

⚠ Dutch notification text (confirmed by client)

Veiligheidscontrole: controleer op contra-indicerende comorbiditeiten voordat je dit activeert — dit plan kan meer dan 45% vet bevatten.

4. Change 3 — Carbohydrates (Frontend — One Exception Only)

⚠ CRITICAL CORRECTION FROM v1

The carb field does NOT become read-only. The dropdown remains unchanged for all profiles. The only change is ONE exception: vegan patients at 2,400-2,800 kcal. For that profile only: 30% is greyed out and 32.5% becomes the lowest selectable value. All other profiles: carb dropdown is completely unchanged.

4.1 The single carb change

Profile	Current behaviour	New behaviour
All omnivore (all kcal)	30% is lowest selectable	UNCHANGED
All vegetarian (all kcal)	30% is lowest selectable	UNCHANGED
Vegan, 1,200 – 2,300 kcal	30% is lowest selectable	UNCHANGED
Vegan, 2,400 – 2,800 kcal	30% is lowest selectable	30% greyed out. 32.5% is now lowest selectable.

Rationale: at 22.5% protein floor + 30% carbs, implied fats = 47.5% which exceeds the 45% ceiling. Raising the carb minimum to 32.5% for this profile keeps fats at 45% exactly (22.5% + 32.5% + 45% = 100%).

i FRONTEND ONLY

No backend SP change required for carbs. This is a single conditional in the frontend dropdown logic.
Condition: IF IsVegan = true AND EnergyTarget >= 2400 THEN grey out 30% in the carb dropdown.

5. Change 4 — Fiber (Backend + Frontend)

 **CRITICAL CORRECTION FROM v1**

The fiber dropdown is NOT a simple floor grey-out. The logic is CUMULATIVE AVAILABILITY. The full dropdown is: 15, 18, 21, 23, 25, 30, 33, 35, 38, 40g (10 values, not 6). Values become available as carb intake increases. Lower band values remain selectable in higher bands. The available set grows with carb grams. Values above the band maximum are greyed out. 50-84g carb rows are documented but NOT active in Stage 1 (keto Stage 2 only).

5.1 Full dropdown values

Replace the current fiber dropdown (25, 33, 38, 40g) with the following 10 values:

Current dropdown	New dropdown
25, 33, 38, 40g	15, 18, 21, 23, 25, 30, 33, 35, 38, 40g

Upper limit: 70g flat across all profiles and all carb bands. The Max target column currently shows a dash — replace with 70g.

5.2 Cumulative availability matrix

The maximum selectable value is determined by the patient's daily carb intake in grams and dietary preference. All values at or below the maximum for that band are selectable. All values above are greyed out. Because the values are cumulative, values from lower bands remain available in higher bands.

Daily carb (g)	Omnivore max	Vegetarian max	Vegan max
50 – 69g †	21g	21g	23g
70 – 84g †	23g	23g	25g

85 – 109g	25g	25g	30g
110 – 124g	30g	30g	35g
125 – 134g	33g	35g	38g
135 – 149g	35g	38g	40g
150 – 199g	38g	40g	40g
≥ 200g	40g	40g	40g
Upper limit (all)	70g	70g	70g

† Rows marked † are Stage 2 (keto) only. Not active in Stage 1. Document now, implement with Stage 2.

5.3 Exact available values per band (for implementation reference)

Carb band	Omnivore selectable	Vegetarian selectable	Vegan selectable
85–109g ★	15,18,21,23,25	15,18,21,23,25	15,18,21,23,25,30
110–124g ★	15,18,21,23,25,30	15,18,21,23,25,30	15,18,21,23,25,30,33,35
125–134g ★	15–23,25,30,33	15–23,25,30,33,35	15–23,25,30,33,35,38
135–149g ★	15–23,25,30,33,35	15–23,25,30,33,35,38	15–23,25,30,33,35,38,40
150–199g ★	15–23,25,30,33,35,38	15–23,25,30,33,35,38,40	15–23,25,30,33,35,38,40
≥200g ★	all (15–40)	all (15–40)	all (15–40)

★ Active in Stage 1. † Stage 2 rows (50–84g carbs) omitted from this table — implement with Stage 2.

5.4 Backend — SetDefaultsOfUserGuidance

Two things are written to UserNutritionsByDietitian for Fiber at profile setup: the interim default value (DailyTarget) and the flat 70g ceiling (MaxDailyTarget). The frontend reads DailyTarget to determine which dropdown values are available for that patient.

i INTERIM VALUES — NOT PERMANENT

These default values are the same across all dietary preferences for a given carb band. In a future release, defaults will become diet-type dependent (separate document from client). Mark this block clearly in code as INTERIM so it is not treated as permanent logic. The dropdown ceiling logic (section 5.2-5.3) is already diet-aware and does not change.

```

-- =====
-- FIBER DEFAULT WRITE (CHG-MACRO-001 v2)
-- INTERIM: same value for all diet types per carb band.
-- Diet-aware defaults to be delivered in a future release.
-- Input: @CarbohydratesMax (carb ceiling grams, already calculated)
-- =====
DECLARE @FiberMin FLOAT;
DECLARE @FiberMax FLOAT = 70.0; -- flat ceiling, all profiles

-- INTERIM: no diet type branching needed here
SET @FiberMin = CASE
    WHEN @CarbohydratesMax <= 69 THEN 15 -- Stage 2 (keto) only, document now
    WHEN @CarbohydratesMax <= 84 THEN 21 -- Stage 2 (keto) only, document now
    WHEN @CarbohydratesMax <= 109 THEN 23
    WHEN @CarbohydratesMax <= 124 THEN 25
    WHEN @CarbohydratesMax <= 199 THEN 30 -- covers 125-134, 135-149, 150-199
    ELSE 33 -- >= 200g
END;

-- Write to UserNutritionsByDietitian WHERE Nutrition = 'Fiber':
-- DailyTarget = @FiberMin <- frontend reads this to set dropdown ceiling
-- MaxDailyTarget = @FiberMax <- displayed as 70g in Max target column

```

Note: @CarbohydratesMax is the carb ceiling grams already calculated earlier in the SP. For most profiles this is $\text{EnergyTarget} \times 30\% \div 4$. For vegan 2,400+ kcal it is $\text{EnergyTarget} \times 32.5\% \div 4$.

5.4a Interim default vs dropdown ceiling — reference table

The default written by the SP (interim, diet-unaware) always sits below the dropdown ceiling (diet-aware). The dietitian always has room to select higher. This table is for developer reference only.

Carb band	Interim default(all diets)	Dropdown maxomnivore	Dropdown maxvegetarian	Dropdown maxvegan	Upperlimit
50–69g †	15g	21g	21g	23g	70g
70–84g †	21g	23g	23g	25g	70g
85–109g	23g	25g	25g	30g	70g
110–124g	25g	30g	30g	35g	70g
125–134g	30g	33g	35g	38g	70g
135–149g	30g	35g	38g	40g	70g
150–199g	30g	38g	40g	40g	70g
≥ 200g	33g	40g	40g	40g	70g

† Stage 2 (keto) rows. Document now, implement with Stage 2. Not active in Stage 1.

5.5 Frontend — fiber dropdown correlation

The fiber dropdown availability is a correlation, not a calculation. The SP has already determined the correct maximum selectable value for that patient and written it to the database. The frontend simply reads it and enables or disables dropdown options accordingly. No band logic or diet-type logic is needed in the frontend.

Item	Detail
Floor source	Read UserNutritionsByDietitian.DailyTarget WHERE Nutrition = 'Fiber'. Already in API response.
Selectable values	All dropdown values $\leq$ @FiberMin are active and clickable
Greyed-out values	All dropdown values $>$ @FiberMin are greyed out (visible, not clickable)
Max display	Show 70g in the Max target column (read from MaxDailyTarget for Fiber row). Replace the current dash.
No frontend calc	Frontend does not calculate the band — it reads the precomputed floor from the DB. Keep it simple.

6. Implementation Order

The order matters. Fats ceiling must go live first. Without it, lowering protein ceilings forces some profiles into an impossible fat residual.

#	Change	Component	Dependency
1	FatsMax → 45% in CaloryBrakets	Database	None — deploy first
2	Fat deviation penalty	Both algorithm SPs	Can run in parallel with 1
3	Protein ceiling dropdown grey-out	Frontend	Needs (1) live first
4	Carb: vegan 2400+ 30% grey-out	Frontend	Independent — can deploy any time
5	Fat dropdown extension + notification	Frontend	Needs (1) live first
6	Fiber SP dynamic floor + 70g ceiling	SetDefaultsOfUserGuidance	Independent — can deploy in parallel with 1-5
7	Fiber dropdown cumulative logic	Frontend	Needs (6) live first

⚠ DO NOT deploy protein or fat frontend changes before CaloryBrakets fats update (step 1) is live.

If fats ceiling stays at 35% while protein ceilings drop, some profiles become mathematically impossible.
 Vegan 1200-2300 kcal: 25% protein + 30% carbs = 45% fats required. If ceiling is 35% the plan breaks.

7. Closure Check — All Profiles Sum to 100%

Profile	Protein ceiling	Carb ceiling	Implied fats	Within 20–45%?
Omnivore 1,200–2,000	35%	30%	35%	✓ Yes
Omnivore 2,100–2,300	32.5%	30%	37.5%	✓ Yes
Omnivore 2,400–2,800	30%	30%	40%	✓ Yes
Vegetarian 1,200–2,300	30%	30%	40%	✓ Yes
Vegetarian 2,400–2,800	27.5%	30%	42.5%	✓ Yes
Vegan 1,200–2,300	25%	30%	45%	✓ At ceiling
Vegan 2,400–2,800	22.5%	32.5%	45%	✓ At ceiling

i NOTE

Both vegan profiles sit exactly at 45%. They have zero slack.
 Any reduction in protein ceiling OR tightening of carb ceiling for vegan profiles would break the closure.
 Stage 2 keto will require a revised fat ceiling — this is why it is deferred.

8. What Does NOT Change

Component	Status
Scoring weights (1.5/15/12/3)	Unchanged. Fat deviation is an additional penalty term, not a weight change.
Meal window percentages	Breakfast 35%, lunch 30-50%, dinner 20-50% — unchanged.
CHG-DEFICIT-002 energy logic	Already deployed and live. Not touched.
Protein default calculation (1.2g/kg)	Not at risk. Covered in a separate document from client. Not in scope here.
Carb dropdown (all non-vegan profiles)	Completely unchanged.

Omega 3/6, Sodium, Potassium	Not in scope for CHG-MACRO-001.
Stage 2 keto parameters	Deferred. No carb floor below 30% in this change.

9. Verification — Expected Outputs After Deployment

Test Case 1 — Vegan, 2,600 kcal, BMI > 25

EnergyTarget = 2,597 kcal (after CHG-DEFICIT-002 deficit). @CarbohydratesMax = $2597 \times 32.5\% \div 4 = 211\text{g}$.

Check	Expected	How to verify
Protein dropdown ceiling	22.5% — values 25,27.5,30,32.5,35 greyed out	Open dietitian panel for patient
Carb dropdown min	32.5% — 30% greyed out	Open dietitian panel for patient
Fiber DailyTarget in DB	40g (carbs 211g, vegan $\geq 200\text{g}$ band)	Query UserNutritionsByDietitian WHERE Nutrition='Fiber'
Fiber MaxDailyTarget in DB	70g	Same query
CaloryBrackets FatsMax	130g (45% of 2600 kcal $\div 9$)	Query CaloryBrackets for 2600 kcal band

Test Case 2 — Omnivore, 1,500 kcal, BMI normal

EnergyTarget = 1,500 kcal (no deficit, BMI < 25). @CarbohydratesMax = $1500 \times 30\% \div 4 = 112.5\text{g}$.

Check	Expected	How to verify
Protein dropdown ceiling	35% — no change, all values selectable	Open dietitian panel
Carb dropdown	Unchanged — 30% is lowest selectable	Open dietitian panel
Fiber DailyTarget in DB	30g (carbs 112g, omni 110-124g band)	Query UserNutritionsByDietitian WHERE Nutrition='Fiber'
Fiber dropdown available	15,18,21,23,25,30 active; 33,35,38,40 greyed	Open dietitian panel
Fiber MaxDailyTarget in DB	70g	Same query

Test Case 3 — Fat notification trigger

Scenario	Expected	Pass if
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Dietitian selects 42.5% fat	Red notification appears immediately	Visible before any other action
Dietitian selects 45% fat	Red notification visible	Visible
Open profile already at 42.5%	Red notification visible on page load	Visible without interaction
Dietitian selects 40% fat	Notification disappears	Not visible at 40%
Save with 42.5% fat	Save completes — notification does not block	Profile saved successfully

10. Developer Checklist

#	Task	Component	Type
1	Update CaloryBrakets FatsMax to 45% grams across all kcal rows	CaloryBrakets table	Database — FIRST
2	Add @ProfileFatPct variable and deviation penalty to scoring	Both algorithm SPs	Backend
3	Add diet flag detection block (@IsVegan/@IsVegetarian)	SetDefaultsOfUserGuidance	Backend
4	Replace static fiber MinFiber with dynamic CASE block	SetDefaultsOfUserGuidance	Backend
5	Write FiberMax = 70g to UserNutritionsByDietitian	SetDefaultsOfUserGuidance	Backend
6	Protein dropdown: grey out values ABOVE ceiling per profile matrix	Frontend	Frontend — after #1
7	Fat dropdown: add 37.5%, 40%, 42.5%, 45%	Frontend	Frontend — after #1
8	Fat notification: red banner at 42.5%+ on select and page load	Frontend	Frontend
9	Carb dropdown: grey out 30% for vegan 2400-2800 kcal only	Frontend	Frontend — independent
10	Fiber dropdown: replace with 10-value list (15-18-21-23-25-30-33-35-38-40g)	Frontend	Frontend
11	Fiber dropdown: cumulative grey-out using FiberMin from DB	Frontend	Frontend — after #4
12	Fiber Max column: display 70g instead of dash	Frontend	Frontend — after #5

 REMINDER

Do not begin frontend changes (items 6-12) before CaloryBrakets fats update (item 1) is live and verified.

Algorithm SP changes (item 2) can run in parallel with item 1.

CHG-DEFICIT-002 must be fully deployed before this change is tested end-to-end.

Client sign-off on CHG-MACRO-001 must be received before development begins.